

A comparison of type theory with set theory

Ansten Klev

Abstract This paper discusses some of the ways in which Martin-Löf type theory differs from set theory. The discussion concentrates on conceptual, rather than technical, differences. It revolves around four topics: sets versus types; syntax; functions; and identity. The difference between sets and types is spelt out as the difference between unified pluralities and kinds, or sorts. A detailed comparison is then offered of the syntax of the two languages. Emphasis is put on the distinction between proposition and judgement, drawn by type theory, but not by set theory. Unlike set theory, type theory treats the notion of function as primitive. It is shown that certain inconveniences pertaining to function application that afflicts the set-theoretical account of functions are thus avoided. Finally, the distinction, drawn in type theory, between judgemental and propositional identity is discussed. It is argued that the criterion of identity for a domain cannot be formulated in terms of propositional identity. It follows that the axiom of extensionality cannot be taken as a statement of the criterion of identity for sets.

Introduction

Set theory and type theory are both said in many ways. Here I shall understand by set theory standard axiomatic set theory, thus ZFC or some variation thereof. By type theory I shall understand Martin-Löf type theory. There are in fact several versions of Martin-Löf type theory, but the differences between these will not play any significant role for us. The only feature I shall assume that is not common to all versions of type theory is a hierarchy of higher types, also called a logical framework. Homotopy type theory is an extension of Martin-Löf type theory obtained by adding the univalence axiom and rules allowing the formation of so-called higher inductive

Ansten Klev
Institute of Philosophy, Czech Academy of Sciences, Jilská 1, 110 00 Prague 1, Czechia
e-mail: klev@flu.cas.cz

types. Most of what I shall say regarding type theory should apply equally well to this extension (which, of course, is not to say that everything one can say regarding Martin-Löf type theory applies equally well to homotopy type theory).

The aim of this contribution is to note some of the ways in which type theory differs from set theory and the predicate logic that it is written in. I hope to make it plausible that the somewhat complex surface structure of type theory hides a very rich conceptual architecture that may be of considerable value to those interested in the philosophical analysis of logical and mathematical notions. I will thus concentrate on the more conceptual aspects of type theory, rather than its more technical aspects. Philosophically inclined logicians and logically inclined philosophers make up the intended audience. It is not presupposed that the reader is an expert in set theory or type theory.

The topics I shall deal with are: sets versus types (section 1); syntax (section 2); functions (section 3); and identity (section 4). Each of these topics provides a lens through which fundamental differences between type theory and set theory may be observed. For the purposes of a comparison of the kind intended here other topics could also have been included. One that springs to mind is axioms (as in set theory) versus rules (as in type theory); but, for reasons of space and time, a study of such further topics will have to wait for another occasion.

1 Sets and types

Set theory is a theory of sets and type theory a theory of types. Both the notion of set and the notion of type may be described as the notion of a class, group, sum, totality, plurality, manifold, etc. of things. The way in which sets are handled in set theory is, moreover, quite similar to the way in which types are handled in type theory. For instance, in set theory we have a set of natural numbers, whose existence is guaranteed by the axiom of infinity; and in type theory a type of natural numbers, whose existence is guaranteed by the rules of natural-numbers formation, introduction, and elimination. In set theory we have an empty set, $\emptyset$, whose existence is either postulated outright or taken to follow from the existence of any other set by application of the axiom of separation; and in type theory a type, often written $\mathbf{0}$, that by definition is empty, namely by its not having any introduction rule. And just as new sets may be formed from old by various set-theoretical operations (union, intersection, pairing, separation, power set), so new types may be formed from old by various type-theoretical operations, indeed operations that often are similar to ones available in set theory (disjoint union of two types, disjoint union of a family of types, generalized Cartesian product).

The fundamental notion of set theory would thus appear to be quite similar to the fundamental notion of type theory. This appearance fades on closer scrutiny: sets

and types are in fact objects of quite different natures. The difference is implicit in Cantor's very first definition of the notion of set (my translation):¹

A manifold (a sum, a set) of elements belonging to some conceptual sphere is called well-defined if, on the basis of its definition and in accordance with the logical principle of the excluded third, it must be regarded as internally determined, both whether any object of that conceptual sphere belongs as an element to the mentioned set, and also whether two objects belonging to the set, in spite of formal differences in the mode of givenness, are equal to each other or not.

Cantor here presupposes a distinction between a set and the conceptual sphere to which the elements of the set belongs. A type can very well be conceived of as a conceptual sphere: a type is a kind, or sort, of thing, whence the things that fall under, or belong to, a type have a certain homogeneity and may therefore be said to constitute a conceptual sphere. Implicit in Cantor's definition is thus the distinction between, on the one hand, a set as a collection of things and, on the other, a type, that is, a kind, or sort, of things.

A similar distinction is quite explicit in another foundational document of set theory. Zermelo (1930), namely, draws a distinction between sets (*Mengen*) and domains of sets (*Mengenbereiche*). A domain of sets in Zermelo's sense is what one today more commonly calls a model of set theory, or a universe of sets. It is this notion of a domain, or universe, of sets rather than that of an individual set that most closely resembles the notion of a type. This understanding of the relation of sets and types is nicely illustrated by Aczel (1978), who provides a model of the constructive set theory CZF—which by the addition of the law of excluded middle yields classical ZF—inside a particular type $\mathbf{V}$ described by Aczel (p. 61) as “a type theoretic reformulation of the classical conception of the cumulative hierarchy of sets”. Thus a universe of sets in the sense of CZF is taken to be a certain type, and an individual set to be an object of this type. De Bruijn (1975) assumes the same understanding of the relation between sets and types in his discussion of how to implement set theory in type theory: he suggests doing so by the introduction of a separate type, SET, and a relation, $\in$, on this type subject to one's favourite axioms of set theory.

Zermelo is careful to distinguish ordinals from sets representing ordinals inside a domain of sets. Thus he seems to assume that besides domains of sets there is also a domain of ordinals. The same assumption is upheld in type theory, where there is in fact an indefinite range of domains of ordinals, namely types usually written $\mathbf{W}(A, B)$, where A is a type of individuals and B a family of types of individuals over A . This illustrates the general point that in type theory each kind of object has its own domain. For instance, a type of ordinals $\mathbf{W}(A, B)$ is not a “subtype” of any other type, in particular not of Aczel's $\mathbf{V}$; nor is the type of natural numbers $\mathbf{N}$ a

¹ Cantor (1882, pp. 114–115): “Eine Mannichfaltigkeit (ein Inbegriff, eine Menge) von Elementen, die irgend welcher Begriffssphäre angehören, nenne ich *wohldefinirt*, wenn auf Grund ihrer Definition und in Folge des logischen Princips vom ausgeschlossenen Dritten es als *intern bestimmt* angesehen werden muss, *sowohl* ob irgend ein derselben Begriffssphäre angehöriges Object zu der gedachten Mannichfaltigkeit als Element gehört oder nicht, *wie auch* ob zwei zur Menge gehörige Objecte, trotz formaler Unterschiede in der Art des Gegebenseins einander gleich sind oder nicht.”

“subtype” of any other type, in particular not of any type of ordinals $\mathbf{W}(A, B)$; nor, finally, is the type $\mathbf{n}$ that by definition has n elements a “subtype” of any other type, in particular not of the type $\mathbf{N}$.

According to another of Cantor’s definitions, a set is “any plurality that can be thought as a one”.² The pair of notions of many and one, or plurality and unity, has a long history in philosophy. For example, Parmenides held that what there is is one; his pupil Zeno argued that there is no plurality; and Anaxagoras countered that what there is is a plurality.³ An example closer to our concerns here is the definition, found in Aristotle (*Metaphysics* XI.9) and Euclid (*Elements* VII Def 2), of a number as a plurality of units or indivisibles. It is natural to think of numbers thus understood as sets of units, as Cantor himself defines numbers.⁴ Indeed, it is natural to regard the pair of notions of plurality and unity as the ancient predecessor of the notion of set: a set is a unified plurality. Such is the understanding of sets assumed not only by the early pioneers of set theory, viz., Cantor and Dedekind (cf. his 1888, art. 2), but also by the, nowadays widely accepted, iterative conception of set (cf. e.g. Linnebo, 2010).

The notion in traditional philosophy that most appropriately may be called a predecessor of the notion of type as understood in type theory is, to my mind, not the notion of plurality, but rather that of genus. According to Aristotle’s definition (*Topics* I.5), the term P is a genus of the term S if P reveals *what* S is, in other words, if P is an appropriate answer to the question of what S is. Aristotle adds a qualification to the effect that S cannot be an individual term. Thus he regards living being as a genus of animal and colour as a genus of white, but not man as a genus of Socrates or white as a genus of a trope of whiteness, such as that found in a piece of paper on my desk. The tradition, especially following Porphyry’s *Isagoge* (cf. Barnes, 2003), speaks of the predicate in these latter cases as a species of its subject. But also a species is taken to be a predicate that reveals a *what* of its subject; hence, for our purposes—which are not exegetical—it seems justified to say that P is a genus of S also when S is an individual term, provided, of course, that P reveals what S is.

It is indeed quite natural to think of the type of an object in type theory as the *what* of that object. What is 5? A natural number. What is $\cos$? A function from $\mathbb{R}$ to $\mathbb{R}$. Any mathematical object is an object of a certain type, and not just an object full stop. In taking this view of mathematical ontology, Martin-Löf type theory differs not only from set theory, but also from type theories with a single type of individuals, usually called ι . Such are, for instance, the simple type theories of Frege (1893) and Church (1940) as well as the ramified type theory of the *Principia Mathematica* (Russell and Whitehead, 1910). Here all individuals are regarded on a par, and natural numbers, for instance, are coded as objects in some already constituted type

² Cantor (1883, p. 587): “Unter einer Mannichfaltigkeit oder Menge verstehe ich nämlich allgemein jedes Viele, welches sich als Eines denken lässt. . .”

³ For these claims and arguments see Kirk et al. (1983, esp. pp. 249–251, 266, 358). For a general overview of the topics of plurality and unity in ancient and mediaeval philosophy, see Meier-Oeser (2001).

⁴ See Cantor (1895, p. 482) and the discussion in Hallett (1984, pp. 119–141).

rather than taken to constitute their own type of individuals. In this respect, therefore, such type theories are more closely related to set theory than to Martin-Löf type theory.⁵

Both “kind” and “sort” are natural English translations of the Greek *genos*; and both have been used in the sense of “type” in logic. Thus, the various sorts, or domains of individuals, in many-sorted logic are just types of individuals;⁶ and in the so-called logical framework of Harper et al. (1993) higher types are called “kinds”. Contemporary philosophy also knows the notion of a sortal concept, which moreover plays a role in philosophical logic (cf. e.g. Lowe, 1989). Russell (1903, § 497), when introducing the term “type” in the relevant sense into logic, could well have chosen “kind”, “sort” or “category”; but perhaps he wanted a term less tainted by the philosophical tradition. The Greek *tupos*, from the verb *tuptō*, to beat, or strike, originally meant the effect of a blow, thus an impression or engraving or the like; but it also meant figure as well as the form or shape of something.⁷ In English, “type” was first used to mean symbol, so that a river might be called a type of human life; but it now also means, among other things: (i) a general form or pattern of a class or kind of things; (ii) such a kind or class itself; (iii) an example—sc., a typical example, an archetype or prototype—of such a class.⁸ Of these different meanings, a type in Russell’s sense, that is, in the sense of type theory, comes closest to (ii). When philosophers contrast types with tokens, and when mathematicians speak of order-types and isomorphism-types, they rather follow sense (i).

Cantor’s first definition of set may be regarded as too restrictive in its requirement that the elements of a set should all be taken from one conceptual sphere. It is, namely, a natural thought that a set can collect together things of arbitrary kinds. There might, for instance, be a set containing the author of this paper, the Mona Lisa, the boolean value *t*, and the empty set. The notion of set may thus be said to be formal in the sense of topic-neutral: a set is not tied to any one topic, any one conceptual sphere, but can have elements from arbitrary conceptual spheres. Types, by contrast, may be thought of as being just these topics or conceptual spheres themselves. In the terminology of Husserl (1913) we may call the notion of set a formal category, and types may be called regions (for more discussion, see Klev, 2017a). This is then yet another articulation of the basic difference between the notion of type and the notion of set that has been emphasized here and that consists in a type’s being a kind, or sort, and a set’s being a collection, namely a collection of objects each of which is of some kind.

⁵ The relation is closer still if the simple hierarchy of types is taken to be a hierarchy of classes; see Quine (1956) and Linnebo and Rayo (2012).

⁶ The idea of a many-sorted language is implicit in Hilbert (1899). An early, perhaps the first, use of the term “many-sorted logic” (or, its German cousin), as well as a definition of the thing itself, can be found in Schmidt (1938). Schmidt (p. 485) emphasizes that he uses “sort” instead of “type” or “kind” so as to avoid the association of levels characteristic of type theory.

⁷ Cf. Liddell & Scott’s *Greek–English Lexicon*, s.v. “τύπος”.

⁸ Cf. *Oxford English Dictionary*, s.v. “type”.

2 Syntax

The syntax of a formal language may seem like a rather technical topic, of more interest to the metamathematician or the programmer than to the philosopher. But the syntax of a formal language is the skeleton of its conceptual architecture, so if we want to understand the latter, we should also need to understand the former. In particular, if we wish to compare conceptual aspects of set theory and type theory, we shall do well in comparing the syntax of the languages they are written in.

2.1 Type structure

Some remarks on type structure are necessary to begin with. I will assume the presence of a hierarchy of higher types in Martin-Löf type theory (cf. Nordström et al., 1990, chs. 19–20). The types in this hierarchy may be divided into three kinds: types of individuals, or individual domains; function types; and a “supertype” **dom** of types, or domains, of individuals. This latter type is often called **set** in the literature, but since we are here using “set” in the sense of set theory, another name seems advisable.⁹

Examples of types of individuals are the type **N** of natural numbers and the type **bool** of boolean values, or truth-values. Being types of individuals, both **N** and **bool** are objects of type **dom**.

It is characteristic of function types in Martin-Löf type theory that they are *dependent*. That a function f is of dependent type means that the type of $f(a)$ depends on the argument a . Thus, for different arguments, a and b , the type of $f(a)$ may differ from the type of $f(b)$.¹⁰ For our purposes here it is, however, enough to consider the simpler case of non-dependent function types. Thus we say that $(A)B$ is a type, namely the type of functions from A to B , whenever A and B are types. The notion of function will be discussed in more detail in section 3 below.

For the development of logic inside type theory a type of propositions is required. In the simple type theory of Church (1940) the type o of propositions is, in the presence of an axiom of extensionality, just the type of truth-values. Truth-values were in fact present already in the logic of the *Grundgesetze* (Frege, 1893), where they served as the meanings (*Bedeutung*) of sentences. For Frege, however, truth-values did not constitute a separate type, but were part of the one grand universe of individuals.¹¹ In Martin-Löf type theory propositions are identified with types

⁹ The use of the term “set” for types of individuals, perhaps inspired by Bishop (1967, p. 13), goes back to (Martin-Löf, 1984). “Domain” was used for the same purpose in (Diller and Troelstra, 1984). One reason why this, rather natural, terminology was not taken up by Martin-Löf himself seems to have been the use of the term “domain” for a very different notion in Dana Scott’s so-called theory of domains (e.g. Scott, 1982).

¹⁰ For a general discussion of dependent types, see Aspinall and Hofmann (2005).

¹¹ In the reconstruction of Frege’s type hierarchy given by Dummett (1973, pp. 44–45) there is a separate type of propositions.

of individuals, in line with the so-called propositions-as-types principle, or Curry–Howard isomorphism.¹² On the one hand, a proposition A may be regarded as a type of individuals, namely as the type of its proofs, or truthmakers. On the other hand, a type of individuals A may be regarded as the proposition that A is inhabited. We may therefore identify the type **dom** of types of individuals with the type **prop** of propositions:

$$\mathbf{prop} = \mathbf{dom}$$

A type of propositions is thus accommodated within the hierarchy of higher types by being identified with the “supertype” **dom** of individual domains.¹³ The resulting conception of proposition is, of course, very different from that assumed in the simple theory of types and predicate logic.

We may also think of the language of set theory as presupposing a certain type structure. Namely, there is a type V of individuals called sets; there is a type **prop** of propositions; and for any types A and B , there is a type $(A)B$ of functions from A to B . How propositions are conceived in this context is of little consequence for us; what matters only is that there is a type of propositions.

The function types introduced above are types of unary functions. Functions of greater arity may be explained by the technique of “Currying”.¹⁴ Note first that if A , B , and C are types, then so is $(B)C$ and therefore also $(A)(B)C$. But a function f of type $(A)(B)C$ may be regarded as a binary function whose first argument is of type A and whose second argument is of type B . Thus, for suitable arguments a and b we may write $f(a, b)$ instead of $f(a)(b)$. Conversely, if f is such a binary function, then for any a of type A , we have a unary function $f(a, _)$ from B to C . Thus we may regard f as a function of type $(A)(B)C$. Similar considerations apply to functions of arity greater than two.

2.2 Terms and types

In the presentation of the syntax of a predicate-logical language it is usual to distinguish terms and formulae. The corresponding distinction in a type theory adhering to the propositions-as-types principle is the distinction between terms and types. Indeed, the propositions-as-types principle is sometimes called the formulae-as-types principle. When comparing the syntax of set theory with the syntax of type theory it is therefore natural to begin by comparing the way in which terms and formulae, respectively types, are formed.

¹² See Howard (1980), Martin-Löf (1984, pp. 5–6, 11–13), de Bruijn (1995), or Wadler (2015).

¹³ In homotopy type theory there is also a more specific conception of propositions. A so-called *mere proposition* is a type of individuals that, intuitively, has at most one element; see The Univalent Foundations Program (2013, ch. 3).

¹⁴ Systematic employment of this technique stems from Schönfinkel (1924, § 2), but the idea is present already in (Frege, 1893, § 36).

In the language of set theory there are no primitive term-forming operators, nor any primitive individual constants; hence, variables are the only terms formable by the primitive vocabulary. The rules for forming formulae follow the usual pattern of predicate logic, with the two basic schemes: (i) function applied to a suitable number of arguments; (ii) variable-binding operator applied to a suitable number of operands.¹⁵ Thus we are thinking of $\in$ and $=$ as binary functions on V , namely of type $(V)(V)\mathbf{prop}$, and of the propositional connectives as functions on $\mathbf{prop}$. The quantifiers are the variable-binding operators of set theory.

It is well known that the application of a variable-binding operator to an operand may be analyzed as a combination of pure variable-binding and function application. For instance, the application of the quantifier $\forall x$ to a formula $A(x)$ may be analyzed as

$$A(x) \dashrightarrow x.A(x) \dashrightarrow \forall(x.A(x)),$$

where $x.A(x)$ is $A(x)$ with the variable x bound. Since we have assumed the presence of a hierarchy of higher types, we may, moreover, identify variable-binding with abstraction, sc., lambda-abstraction or functional abstraction. What is here written $x.A(x)$ may thus be thought of as a function from V into $\mathbf{prop}$, thus of type $(V)\mathbf{prop}$. A more familiar notation for $x.A(x)$ would be $\lambda x.A(x)$, but we shall employ the lambda for a slightly different purpose below. The universal quantifier, finally, may then be regarded as a higher-order function, of type $((V)\mathbf{prop})\mathbf{prop}$, for it takes a function, say $x.A(x)$, and yields a proposition $\forall(x.A(x))$.

Both the terms and the types of type theory are formed by the same two schemes (i) and (ii). In particular, in the version of type theory assumed here there is only one variable-binding operator, namely abstraction.¹⁶ The modes of construction in set theory and type theory, both endowed with a hierarchy of higher types, are thus the same: function applied to a suitable number of arguments and variable-binding in the form of abstraction. The primitive vocabulary available for forming terms and types in Martin-Löf type theory is, however, much richer than the primitive vocabulary of set theory. Indeed, whereas the primitive vocabulary of set theory consists of $\in$ and $=$ together with the logical connectives and the quantifiers, the range of primitive constants in Martin-Löf type theory is indefinite. Martin-Löf type theory is, namely, open in the sense that the objects of type **dom** are not given once and for all, in a list or inductive definition, say. Rather, objects of type **dom** (i.e., types of individuals) and functions with co-domain **dom** may, in effect, be introduced at will, subject only to the requirement that the types thus introduced or formable satisfy the general description of what a type of individuals is (cf. Martin-Löf, 1984, pp. 7–8). In general, this requirement is met by providing any newly postulated type of individuals A or function Φ forming such types with one or more introduction rules. Once introduction rules for A or Φ are provided, an elimination rule may be formulated,¹⁷ whereby further pieces of primitive vocabulary are introduced.

¹⁵ For the distinction between function and operator, see Church (1956, §§ 3, 6).

¹⁶ Dependent function types introduce a further variable-binding operator; thus with dependent function types present, there are two variable-binding operators.

¹⁷ Dybjer (1994, 2000) provides a general scheme for the formulation of such rules.

The openness of the primitive vocabulary of type theory may be illustrated by the following, somewhat perverse, example. In order to do arithmetic in type theory one usually introduces a type of natural numbers, $\mathbf{N}$, and three associated constants: the individual constant 0, the successor function, s , and a recursor R , which makes possible definition by recursion and proof by induction on $\mathbf{N}$ (e.g. Martin-Löf, 1984, pp. 71–76). Assuming all of this to have been introduced, there is, however, nothing stopping one from introducing further primitive vocabulary, $\mathbf{N}^*$, 0^* , s^* , and R^* , satisfying rules structurally identical to those governing the corresponding non-starred constants. In other words, there is nothing stopping one from introducing a duplicate of the type of natural numbers and its associated constants. There might not be much use in such duplication; but the question of usefulness is besides the point here, where we just want to illustrate the openness of the primitive vocabulary of type theory. One can of course exploit this openness in more inventive ways than mere duplication of well-known types and their associated constants. In fact, inductively defined types quite generally are to be formalizable in type theory; and any such formalization introduces new primitive vocabulary.

2.3 Judgements

Besides term and type, there is in type theory a further logico-grammatical category: judgement.¹⁸ At the most general level, there are two forms of (categorical) judgement in type theory:

$$\begin{aligned} a &: \mathcal{C} \\ a = b &: \mathcal{C} \end{aligned}$$

The first may be read as saying that a is an object of category $\mathcal{C}$ and the second as saying that a and b are identical objects of category $\mathcal{C}$. Here “category” is used not quite as in “logico-grammatical category”, but in a more ontological sense. This use of the term will seem quite natural if (i) we analyze $a : \mathcal{C}$ as being of the form subject–copula–predicate, and (ii) keep in mind that one of the meanings of the Greek *katēgoria* is “predicate”. The specific forms of judgement assumed in any particular version of type theory depends on which categories are assumed. In the version of type theory assumed in this paper the categories are **type** and A , for any **type** A . Thus we get the following forms of (categorical) judgement:

$$\begin{aligned} A &: \mathbf{type} \\ A = B &: \mathbf{type} \end{aligned}$$

and for any **type** A ,

¹⁸ The term “judgement” in this sense was introduced by Martin-Löf (1982). The term had of course been used in logic before, but it had fallen out of fashion during, say, the first decades of the 20th century; see e.g. Carnap (1934, § 1). For the early history of the use of the term in logic, see Martin-Löf (2011); for aspects of the later history, see Sundholm (2009).

$$\begin{aligned} a &: A \\ a = b &: A \end{aligned}$$

Examples of judgements are thus

$$\begin{aligned} \mathbf{N} &: \mathbf{type} \\ \mathbf{t} &: \mathbf{bool} \\ \mathbf{s} &: (\mathbf{N})\mathbf{N} \\ \mathbf{s}(2+1) &= 2 + \mathbf{s}(1) : \mathbf{N} \end{aligned}$$

It is essential for anyone wishing to understand the relation between the language of set theory and the language of type theory to appreciate the difference between the type theorist's $a : A$ and the set theorist's $a \in b$. The first says that a is an object of the type, or kind, or sort A , whereas the latter says that, within the universe, or type, of sets, the relation $\in$ obtains between a and b . Instances of the form of judgement $a : A$ in the type structure of set theory are

$$\begin{aligned} \emptyset &: V \\ \{\emptyset, \{\emptyset\}\} &: V \\ \in &: (V)(V)\mathbf{prop} \\ \mathbf{Ord} &: (V)\mathbf{prop}, \end{aligned}$$

where $\mathbf{Ord}(a)$, for $a : V$, is the proposition that a is a Von Neumann ordinal. The first two of these judgements says that $\emptyset$, respectively $\{\emptyset, \{\emptyset\}\}$, is a set; the third that $\in$ is a binary propositional function; and the fourth that $\mathbf{Ord}$ is a unary propositional function on the universe of sets V . These are things that cannot be expressed in the language of set theory, in particular not by a formula of the form $a \in b$. That $\emptyset$ is a set, for instance, cannot be expressed by a set-theoretical formula. In fact, reflection on type predications—assertions that such and such is an object of some type—shows that they cannot be taken to have the form of predicate-logical formulae (cf. Klev, 2018a). We are not forced to conclude, as Wittgenstein (1922, esp. 4.126) seems to have done, that type predications cannot be expressed at all, that the type of an object a is something that cannot be said, but only shown in our use of a symbol for a . One can instead introduce—and explain the meaning of—a new form of statement for the expression of type predications. The judgement $a : A$ is such a novel form of statement.

When a type A is regarded as a proposition, an object a of type A may be regarded as a proof, or a truthmaker, of A .¹⁹ The judgement $a : A$ is thus to be read as saying that a is a proof of the proposition A .²⁰ It should be emphasized that $a : A$ is not itself a proposition. In fact, the logico-grammatical categories of judgement and proposition are disjoint: no judgement is a proposition, nor is any proposition a

¹⁹ For the relation between, on the one hand, recent discussions on truthmakers and truthmaker semantics and, on the other, Martin-Löf type theory, see Sundholm (1994) and Klev (2017b).

²⁰ In the literature, also the term “proof-object” is used, following Diller and Troelstra (1984).

judgement. Thus, iterated colons, as in “ $b : (a : A)$ ”, make no sense in type theory.²¹ Nor is there any need for such iterated colons, since the relation between a term and its type is taken to be internal and not in need of proof: if a is of type A , then its being so can be seen either immediately from the form of a or else from the form of what results after a sequence of simple computation steps. This point is closely related to “the basic intuitionistic idealization that we can recognize a proof when we see one” (Kreisel, 1962, p. 202) and to the requirement on a type theory that the relation between a term and its type be decidable in the recursion-theoretic sense.

The theorems of type theory are judgements and not propositions. Thus it is not

$$A$$

but

$$a : A$$

that is the form of a theorem of type theory (albeit not the only form). The logical form of a theorem must be the form of something that is assertion-apt, in the sense of being suitable as the object of an act of assertion. Propositions—be they conceived as types of proofs, as truth-values, or as sets of possible worlds, say—are not assertion-apt in this sense. The judgement that a is a proof of A , by contrast, is assertion-apt and thus has the logical form appropriate for a theorem.

If we can demonstrate that a proposition A has a proof, or a truthmaker, then we are certainly justified in asserting that A is true. Thus it makes good sense to introduce a new form of judgement A *true* governed by the rule

$$\frac{a : A}{A \text{ true}}$$

It must be emphasized that the *true* occurring here is not a truth-predicate in the ordinary sense. A predicate in the usual sense always yields a proposition (or formula) when applied to a suitable number of arguments, but A *true* is a judgement and not a proposition. This *true* should rather be compared to the sign $\vdash$ of judgement, or assertion, in Frege’s and Russell’s systems:

$$\vdash A$$

This assertion sign, the so-called turnstile, is not used in forming new propositions, but in forming a judgement from a proposition. Thus, unlike the propositional connectives and the quantifiers, the assertion sign turns a proposition into something assertion-apt. Employing grammatical terminology, we may say that the assertion sign supplies the mood of the assertion $\vdash A$, whereas the proposition A is the content. A content by itself is not assertion-apt; what is required for assertion is a composite of content and mood, more specifically, of content and assertoric mood.

²¹ This contrasts with so-called justification logic (cf. Artemov and Fitting, 2016), where $a : A$, understood as “ a is a justification of A ”, is a proposition. Thus, for instance $b : (a : A)$ and $(a : A) \supset A$, are well-formed formulae there.

In the language of set theory, as in predicate-logical languages more generally, no distinction similar to that between proposition and judgement is drawn. In particular, the theorems of set theory are not regarded as being of a logico-grammatical category different from that of formulae. The string of symbols $\vdash A$ can of course be found in presentations of set theory, but it is then regarded as expressing a meta-mathematical property of A , namely that there exists a sequence of formulae coding a derivation of the formula A . We may regard this metamathematical “ $\vdash A$ ” as a judgement, but only as a judgement in the metalanguage of set theory (or any other formal theory) and not as a judgement of set theory proper.²² Inside set theory we must instead regard the formulae themselves as assertion-apt; no additional element such as the assertion sign is required in order to turn such a formula into something of the form of a theorem. If we take assertions to have content–mood structure, we must therefore regard the formulae of set theory as coming equipped with the assertoric mood. But then we shall be forced to hold that the antecedent A of an implication $A \supset B$ as well as any of the disjuncts of a disjunction $A \vee B$ have assertoric mood, which they clearly do not have.²³ For the purposes of conceptual clarity it is therefore quite useful to draw a distinction such as that between proposition and judgement.

2.4 Hypothetical judgements

In the simple theory of types, as well as in higher-order logic and many-sorted logic, variables are typed globally, in the sense that the type of a variable is specified once and for all by the formation rules of the language. In Martin-Löf type theory, by contrast, variables are typed locally, namely in a so-called context

$$x_1 : A_1, \dots, x_n : A_n,$$

where the A_i ’s are types. If there are free variables in a judgement $a : \mathcal{C}$ or $a = b : \mathcal{C}$, then such a judgement therefore does not have a determinate meaning unless a context is provided in which the types of these variables are declared. The general form of judgement in type theory must therefore also list the context. A common notation uses the turnstile, as follows:

$$\begin{aligned} x_1 : A_1, \dots, x_n : A_n &\vdash a : \mathcal{C} \\ x_1 : A_1, \dots, x_n : A_n &\vdash a = b : \mathcal{C} \end{aligned}$$

These judgements should be read as saying that a is an object of category $\mathcal{C}$, respectively that a and b are identical objects of category $\mathcal{C}$, in the context $x_1 : A_1, \dots, x_n : A_n$. This in turn may be recast as a *hypothetical* judgement: a is an object of category

²² See Curry (1963, ch. 2.C) for such an account of the metamathematical “ $\vdash A$ ”.

²³ This is sometimes called the “Frege point” after Geach (1972, pp. 254–270); see e.g. Frege (1983, pp. 201–203).

$\mathcal{C}$, or a and b are identical objects of category $\mathcal{C}$, provided $x_1 : A_1, \dots, x_n : A_n$. We get a more familiar form of hypothetical judgement by letting $\mathcal{C}$ be a proposition A and by thinking of the type assignment $x_i : A_i$ as the assumption that A_i is true:

$$A_1 \text{ true}, \dots, A_n \text{ true} \vdash A \text{ true}$$

This is the judgement that A is true provided, or on the assumption, or condition, that $A_1, \dots, A_n$ are all true.

Hypothetical judgement is another novelty of type theory in comparison with set theory and ordinary predicate logic. In particular, hypothetical judgements should not be confused with implications, that is, propositions of the form $A \supset B$, for a hypothetical judgement is not a proposition (cf. Sundholm, 2012). A hypothetical judgement should rather be compared to a sequent in the sense of Gentzen (1936, § 5), namely a sequent

$$A_1, \dots, A_n \longrightarrow A$$

to be read as saying that A is true on the assumption that $A_1, \dots, A_n$ are all true and governed by introduction and elimination rules together with structural rules (these sequents are therefore not quite those of ordinary sequent calculi, which have so-called left and right rules rather than introduction and elimination rules). This, in turn, may be understood as a gloss on the assertion that the relation of consequence holds between the propositions $A_1, \dots, A_n$ and A .

3 Functions

Among the fundamental notions of mathematics one finds not only the notion of a domain or collection of objects, but also that of an operation on, or transformation, or mapping, of objects of such a domain, what is usually called a function. We have already come across (non-dependent) function types: if A and B are types, then so is $(A)B$. In Martin-Löf type theory we take this type to be explained by the following two elimination-like rules:

$$\frac{f : (A)B \quad a : A}{f(a) : B} \quad \frac{f : (A)B \quad a = a' : A}{f(a) = f(a') : B}.$$

The first rule says that whenever $f : (A)B$ and $a : A$, then there is an object $f(a)$ of type B , namely the result of applying f to a . With the function type $(A)B$ there is thus associated a primitive notion of application to an argument. The second rule says that a function applied to identical arguments yield identical results.

In set theory it is not common to regard the notion of function as a primitive notion. On the contrary, one there usually takes the notion of function to be explained in terms of the primitive notions of set theory. More precisely, a function f is there defined to be a set f of ordered pairs satisfying the condition that if $\langle x, y \rangle \in f$ and

$\langle x, y' \rangle \in f$ are both true, then so is $y = y'$.²⁴ Inside set theory, there might thus be no need to talk about functions in addition to sets. If we are to analyze the language and conceptual architecture of set theory, we do, however, still need to invoke functions, as we have already done when sketching a type structure for set theory. In particular, the relations of elementhood, $\in$, and identity, $=$, will then be analyzed as binary propositional functions on the universe V of sets, as expressed by the following judgements:

$$\begin{aligned}\in & : (V)(V)\mathbf{prop} \\ = & : (V)(V)\mathbf{prop}\end{aligned}$$

Yet another function is presupposed by the definition of the application of a function *qua* set of ordered pairs to an argument. Namely, in order to define the term $f(a)$, where both f and a are sets, the syntax of predicate logic requires that there be some function connecting f and a . Thus, we must think of $_{-1}(_2)$ as a binary function on the universe V , as follows:

$$_{-1}(_2) : (V)(V)V$$

Of course, one could also use some other notation here, such as a left quotation mark, thus writing $f'a$, as in Shoenfield (1967, p. 245).

Since the primitive language of set theory contains no term-forming operators, the definition of any n -ary function symbol F must there take the form

$$Fa_1 \dots a_n = b \leftrightarrow_{\text{def}} \varphi[a_1, \dots, a_n, b] \quad (\text{Def-}F)$$

Here $\varphi[a_1, \dots, a_n, b]$ is a formula whose free variables are among $a_1, \dots, a_n, b$ such that

$$\forall x_1 \dots x_n \exists! y \varphi(x_1, \dots, x_n, y)$$

is a theorem. The application function, in particular, may be introduced by the definition

$$f(a) = b \leftrightarrow_{\text{def}} (\psi[f, a] \wedge \langle a, b \rangle \in f) \vee (\neg \psi[f, a] \wedge b = \emptyset),$$

where $\psi[f, a]$ is a formula expressing that f is a function in the sense of set theory and that there is x such that $\langle a, x \rangle \in f$. Notice, however, that the *definiendum* here is not $f(a)$, but rather $f(a) = b$. More generally, the *definiendum* in the definition (Def- F) is not the term $Fa_1 \dots a_n$, but rather the formula $Fa_1 \dots a_n = b$. What is explained by such a definition is, therefore, not the term, but the formula. The term $Fa_1 \dots a_n$ is thus rendered an improper symbol in the sense of Russell and Whitehead (1910, p. 66), viz., a symbol that has no meaning in isolation, but only in certain contexts. The term $f(a)$, in particular, has no meaning outside the context of formulae of the form $f(a) = b$ or formulae defined from such formulae. The proposition that $2 + 2$ is an even number, for instance, cannot therefore be analyzed simply as $\text{Even}(2 + 2)$, but must be analyzed as $\exists n(2 + 2 = n \wedge \text{Even}(n))$.

²⁴ Hausdorff (1914, p. 33) appears to have been the first to define the notion of function in this way solely in terms of set theory. On Hausdorff's definition in a historical context, see Ulrich Felgner's article in (Hausdorff, 2002, pp. 621–633).

Functions *qua* sets of ordered pairs should thus not be compared to objects of function type $(A)B$. They may, however, be compared to objects of type $A \rightarrow B$. In Martin-Löf type theory $A \rightarrow B$ is a type of objects that, just as the relevant sets of ordered pairs, are individuals coding functions. It can be formed whenever A and B are types of individuals:

$$\frac{A, B : \mathbf{dom}}{A \rightarrow B : \mathbf{dom}}$$

There is a more general form of type, written $\Pi(A, B)$, of individuals coding dependent functions; but, as elsewhere in this article, we may concentrate on the simpler non-dependent case, $A \rightarrow B$. The introduction rule for $A \rightarrow B$ allows us to form an individual $\lambda(f)$ for any $f : (A)B$ when A and B are types of individuals:

$$\frac{A, B : \mathbf{dom} \quad f : (A)B}{\lambda(f) : A \rightarrow B}$$

Just as in set theory, we need an application function to account for the application of an element $c : A \rightarrow B$ to an argument. Such an application function is provided by the elimination rule for $A \rightarrow B$:

$$\frac{c : A \rightarrow B \quad a : A}{\mathbf{ap}(c, a) : B}$$

The application function $\mathbf{ap}$ is defined by laying down that it is to satisfy the following equation:

$$\mathbf{ap}(\lambda(f), a) = f(a) : B$$

From this equation it is clear that $\lambda(f)$ may indeed be thought of as a code of $f : (A)B$.

Those familiar with Frege's logical system in the *Grundgesetze* may recognize the similarity between the terms $\lambda(f)$ and Frege's courses-of-values (*Wertverläufe*; Frege, 1893, § 9). A course-of-values $\dot{\alpha}.f(\alpha)$ can be formed whenever f is a function from individuals to individuals (or, from objects to objects, in Frege's terminology). Moreover, there is a binary application function, $-_1 \frown -_2$, defined on individuals, satisfying

$$\dot{\alpha}.f(\alpha) \frown a = f(a)$$

The rules governing courses-of-values are thus quite similar to those governing $\lambda(f)$. An essential difference, however, is that a Fregean course-of-values is of the same type as any argument to which it may be applied via $-_1 \frown -_2$. Self-application therefore makes sense, which is one of the reasons (though not the only reason) for the inconsistency of Frege's system. In Martin-Löf type theory, by contrast, the object $\lambda(f)$ is of type $A \rightarrow B$, whereas any object to which $\lambda(f)$ may be applied via $\mathbf{ap}$ is of type A .

We should emphasize the difference between the primitive notion of application associated with the function type $(A)B$ and application via an application function such as $\mathbf{ap}$, $-_1 \frown -_2$ or $_{-1}(_2)$. Consider, for instance, the term $\mathbf{ap}(c, a)$. It involves both notions of application: the primitive application of $\mathbf{ap}$ to c and a and the appli-

cation of c to a via the function ap . It is clear that some primitive notion of function application is needed. As Frege (1892, p. 205) and Bradley (1893, ch. 3) in effect noted, an infinite regress soon arises if we try to get rid of the primitive notion of function application in favour of application via an application function. For instance, if we invoke an application function ap^* to account for the application of ap to c and a , as in $\text{ap}^*(\text{ap}, c, a)$, then we either need to say that ap^* is applied primitively to its arguments or else invoke yet a further application function and thus be on the brink of regress.

4 Identity

Recall that in type theory we have two basic forms of (categorical) judgement: $a : \mathcal{C}$ and $a = b : \mathcal{C}$. Just as there is a fundamental difference between the formula $a : \mathcal{C}$ of type theory and the formula $a \in b$ of set theory, so there is a fundamental difference between the formula $a = b : \mathcal{C}$ of type theory and the formula $a = b$ of set theory or of predicate logic quite generally. The most fundamental difference is in logico-grammatical category: whereas the former is a judgement, the latter is a proposition. In particular, a formula $a = b$ of predicate logic is to be analyzed as the result of applying a certain binary propositional function, $= : (A)(A)\mathbf{prop}$, to objects $a, b : A$ of the underlying domain A . The judgement $a = b : \mathcal{C}$, by contrast, does not have function–argument structure, nor is it a proposition. As a consequence, the propositional connectives and the quantifiers are not applicable to identity judgements as they are to identity propositions.

The wish to do predicate logic in type theory very nearly forces one to introduce a propositional function **Id** such that for any type of individuals A and any $a, b : A$, the result **Id**(A, a, b) of applying **Id** to A , a , and b is the proposition that a and b are identical. Thus, we are to have the following formation rule:

$$\frac{A : \mathbf{dom} \quad a : A \quad b : A}{\mathbf{Id}(A, a, b) : \mathbf{prop}}$$

That is, whenever A is a type of individuals and a and b are elements of A , then there is a proposition **Id**(A, a, b). Because of the identification $\mathbf{prop} = \mathbf{dom}$, we could also have written **Id**(A, a, b) : **dom** in the conclusion, but **prop** is more suggestive in this case. The formation rule for **Id** does of course not by itself make **Id**(A, a, b) into an identity proposition. But by supplying the function **Id** thus postulated with the introduction rule

$$\frac{a : A}{\text{refl}(a) : \mathbf{Id}(A, a, a)}$$

and a corresponding elimination rule, it can be seen that **Id**(A) indeed behaves as identity on A . In particular, it follows easily from these rules that **Id**(A) is an equivalence relation such that if **Id**(A, a, b) is true, then a and b are indiscernible by unary propositional functions over A (cf. e.g. Nordström et al., 1990, pp. 57–59). Type the-

ory thus recognizes both judgemental identity, $a = b : A$, and propositional identity, $\mathbf{Id}(A, a, b)$.

Identity types, $\mathbf{Id}(A, a, b)$, are central in homotopy type theory and have been the subject of many important metamathematical studies.²⁵ The relation between judgemental and propositional identity, and the treatment of identity more generally, is, moreover, one of the more intriguing philosophical aspects of Martin-Löf type theory. Here I will restrict myself to one topic, namely the conceptual priority of judgemental identity over propositional identity. Again we shall see how type theory allows a much deeper analysis of logical notions—in this case the notion of identity—than anything set theory and predicate logic can offer.

In order to have the right to take f to be a function on some domain A , we must presuppose that A has been equipped with a notion of identity. For we may take f to be a function on A only if we know that f is well-defined, that is, only if we know that the results $f(a)$ and $f(b)$ of applying f to identical arguments $a, b : A$ are identical; or, in symbols:

$$a = b \implies f(a) = f(b) \quad (\text{Well-def})$$

The propositional function of identity on A , namely the function $= : (A)(A)\mathbf{prop}$, is a binary function on A . Being such a function, it therefore presupposes that A comes equipped with a notion of identity. On pain of circularity, this notion of identity, presupposed by the propositional function of identity, cannot itself be propositional identity. In particular, the identity, $=$, featuring in (Well-def) cannot be propositional identity. Thus the notion of a function f on a domain A presupposes some notion of identity other than propositional identity.

Judgemental identity, since it is not a propositional function on A , may well serve as the notion of identity presupposed by the notion of a function on A . It does serve this role in Martin-Löf type theory. The formulation there of (Well-def) is, indeed, one of the primitive, meaning-giving rules governing functions, and invokes only judgemental identity (see section 3):

$$\frac{f : (A)B \quad a = b : A}{f(a) = f(b) : B}$$

The formulation of (Well-def) in terms of propositional identity $\mathbf{Id}$, by contrast, is a theorem, easily established by the rule of $\mathbf{Id}$ -elimination. We may conclude that judgemental identity is conceptually prior to propositional identity, since the functionhood of propositional identity presupposes judgemental identity. It is worthwhile noting that this argument relies only on the difference in logico-grammatical category between judgemental and propositional identity, and not on the rules that govern these two notions of identity (which tend to vary between various versions of type theory).

Another way of seeing the priority of judgemental identity over propositional identity follows the idea that the criterion of identity for a type A cannot be expressed by propositional identity, though it may be expressed by judgemental iden-

²⁵ Examples are Streicher (1993), Hofmann (1997), and Hofmann and Streicher (1998).

tity. It is a general requirement on a category $\mathcal{C}$ that it be determinate what it is for objects $a, b : \mathcal{C}$ to be identical. A good example is afforded by the rational numbers. Someone who does not know that different fractions may be the same rational number, such as are $\frac{1}{2}$ and $\frac{2}{4}$, cannot be said to have understood the concept of a rational number. Or, in a more ontological turn of phrase: it is part of the essence of rational numbers that fractions $\frac{m}{n}$ and $\frac{m'}{n'}$ such that $mn' = m'n$ are the same rational number, namely they are the same “in spite of formal differences in the mode of givenness”, to quote Cantor (see the quotation in section 1).²⁶ The explanation of a type A , being a category, must therefore include an explanation of what it is for objects $a, b : A$ to be the same. Or, again in a more ontological terminology: the essence of a type A must include a criterion of identity, namely a criterion that determines what it is for objects $a, b : A$ to be identical. It can then be seen by reasoning similar to that above that the criterion of identity for A cannot be expressed in terms of the propositional function of identity on A (cf. Klev, 2018b). This function, namely, presupposes that A has already been equipped with a notion of identity; but it is the criterion of identity for A that is to determine what it is for objects $a, b : A$ to be identical.

It follows that we cannot regard the axiom of extensionality in set theory,

$$a = b \leftrightarrow \forall x(x \in a \leftrightarrow x \in b),$$

as a formulation of the criterion of identity for sets. Just as all other axioms of set theory, this is an ordinary predicate-logical formula, so the identity sign occurring in it must be taken to signify a propositional function on the universe V of sets, and therefore to presuppose a criterion of identity for V . Or, again, the well-formedness of the axiom of extensionality presupposes that V has been equipped with some notion of identity; hence it cannot be by means of this axiom that V is so equipped.

The identity judgement $a = b : V$, by contrast, lends itself to the expression of the identity criterion for V , since it does not presuppose any notion of identity on V . Indeed, it is by means of judgements $a = b : \mathcal{C}$ that the identity criterion for a category $\mathcal{C}$ in Martin-Löf type theory is stated. For instance, the criterion of identity of the function type $(A)B$ may be taken to be contained in the following two rules:

$$\frac{f = g : (A)B \quad a : A}{f(a) = g(a) : B} \quad \frac{f, g : (A)B \quad x : A \vdash f(x) = g(x) : B}{f = g : (A)B}$$

Thus, we are to have $f = g : (A)B$ if and only if $f(a) = g(a) : B$ for arbitrary $a : A$. The criterion of identity for the function type $(A)B$ is thus in a sense reduced to the criterion of identity for B . The criterion of identity for a type A of individuals follows from a stipulation, stated in terms of judgemental identity, of when two so-called canonical objects of A are identical; we shall not go into the details here (cf. Martin-Löf, 1984, pp. 7–10).

²⁶ Cantor’s use of the term “mode of givenness” (*Art des Gegebenseins*) might have inspired Frege’s use of the same term in his elucidation of the notion of sense; see Sundholm (2001, pp. 65–67).

5 Concluding remarks

A theory that is put to service as a foundation for mathematics provides mathematics not only with a theoretical foundation, that is, a set of basic principles (axioms or rules, say) on the basis of which the theorems of mathematics are to be demonstrable: it also provides it with a certain conceptual architecture, or an “ideology”, an articulation of its basic concepts and their interrelations. In the foregoing pages I have contrasted the two ideologies thus provided by set theory and type theory. We have seen, for instance, how the two primitive notions of set and type differ from each other, as well as how type theory differs from set theory in taking the notion of function to be primitive. Besides outlining such differences, I have tried to convey how rich and nicely detailed the conceptual architecture of type theory is. On first sight such intricacy may seem superfluous. Do we, for instance, need a distinction between judgements and propositions or a distinction between judgemental and propositional identity? Set theory draws no such distinctions, but it has nonetheless been quite successful as a foundation for mathematics. Indeed, the ideological simplicity of set theory is certainly part of the explanation of its success. I have tried to show how the subtle distinctions offered by type theory can be of use in the philosophical analysis of logical and mathematical notions, namely as tools that allow one to dig deeper where set-theoretical ideology gives in.

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